Hyperspectral Image Classification Framework Based on Multichannel Graph Convolutional Networks and Class-Guided Attention Mechanism

Hao Feng, Yongcheng Wang, *Member, IEEE*, Chi Chen, Dongdong Xu, Zhikang Zhao, and Tianqi Zhao

***Abstract*— Graph convolutional networks (GCNs) can extract features of samples in non-Euclidean space, which can be used for hyperspectral image (HSI) classification in collaboration with convolutional neural networks (CNNs). The features of GCNs and CNNs are incompatible to a certain extent, and traditional graph convolution methods use a single-channel 2-D matrix to extract features. As a result, it is difficult to explore the relationships fully and flexibly between samples. To further exploit the potential of these two networks for collaborative extraction of HSI features, we propose a fusion framework based on multichannel GCNs and class-guided attention mechanism (MG^{2}A). Specifically, a multichannel GCN (MGCN) module is designed for batchwise network training, where the adjacent matrix of each channel contains different information between samples. In addition, we develop a class-guided attention mechanism to adaptively fuse the features of multiple MGCN modules and learn the transformation process of the features. Finally, the features of CNNs and MGCN modules are fused at multiple layers through a fusion framework. Experimental results on four benchmark HSI datasets show that MG^{2}A achieves better classification performance compared to other state-of-the-art methods.**

***Index Terms*— Deep learning, feature fusion, graph convolutional networks (GCNs), hyperspectral image (HSI) classification.**

I. Introduction

IN EARTH observation techniques, hyperspectral imaging can capture tens or even hundreds of contiguous spectral bands. This rich spectral information can be used to mine spectral features beyond the visible light range, providing more detailed data support for subsequent land-cover analysis tasks. In recent years, hyperspectral remote sensing image classification techniques have been widely used in various fields, such as environmental detection [1], urban development and planning [2], and precision agriculture [3].

In the hyperspectral remote sensing image classification task, each pixel is assigned a unique semantic label. In early studies of hyperspectral image (HSI) classification techniques, most studies often focused on extracting spectral features. Examples are k-nearest neighbors [4], support vector machines (SVMs) [5], and random forests [6]. It is difficult for these models to extract deeper features in high-dimensional spectral information. Moreover, most models take spectral vectors as input, which makes it more difficult to explore spatial contextual information, resulting in salt-and-pepper noise [7] in the classification results. With the continuous development of hardware technologies, deep learning is widely used in remote sensing image processing tasks [8] due to its powerful deep feature extraction capability. Common deep learning models include convolutional neural networks (CNNs) and stacked autoencoders (SAE). Chen et al. [9] first introduced a deep learning algorithm to the field of HSI classification, employing SAE and principal component analysis (PCA) techniques. Mou and Zhu [10] proposed a CNN based on the HSI band attention mechanism. The attention vector of the model is designed to boost the weights of bands with richer features and suppress the weights of bands with weaker features. Inspired by RGB image classification, some models use 2D-CNNs for HSI classification [11], [12], which extract spatial–spectral information by constructing samples in the form of patches. However, 2D-CNNs will compress high-dimensional band or channel information during PCA or convolution, which will affect the model's ability to extract spectral features. Therefore, the ability of the network to extract spectral–spatial features needs to be considered. Some models used 1D-CNNs and 2D-CNNs to jointly extract spectral and spectral–spatial features of HSIs, respectively [13], [14]. Ma et al. [15] designed a double-branch dual-attention (DBDA) mechanism network that includes a spectral feature extraction branch and a spatial feature extraction branch. In addition, a growing number of studies have started to focus on 3D-CNNs [16], [17], [18]. The 3-D convolution kernel is shifted in the three dimensions of the patch for weighted summation, which can extract both spatial and spectral features. Li et al. [19] proposed an end-to-end spectral–spatial CNN (SSCNN). This model achieves better classification performance than 2D-CNNs with fewer learnable parameters. Roy et al. [20] proposed an attention-based adaptive spectral–spatial kernel residual network (A^{2}S^{2}K). Specifically, two types of 3-D convolution kernels with different sizes are used to extract spectral and spatial features, respectively. Then, the attention mechanism is constructed to perform feature fusion. Finally, the classification results

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are obtained by combining the residual networks. To fully exploit the spatial information of samples and feature maps, multiscale fusion networks and pyramid networks have also been used in the field of HSI classification. Paoletti et al. [21] proposed a deep pyramidal residual network (P-RESNET). In addition, transformer-based networks [22] have also shown competitive performance. Mei et al. [23] restricted the multihead self-attention to the local spatial–spectral background by constructing a grouped pixel embedding.

Despite the strong feature extraction capability of CNNs, it is difficult to obtain information between samples, especially for samples with large mutual spatial distances. It is important to note that the distribution of ground-based objects of various types in the HSI is relatively concentrated, and distances or spectral differences between samples of the same class and samples of other classes will be more stable. When constructing the adjacency matrix of graph samples, the spatial distance information or spectral difference between samples can be encoded as edge information of the graph. Moreover, GCNs can efficiently extract sample features from the non-Euclidean distribution of HSI. Especially, in the adjacency matrix construction, the latent information of HSIs is further mined by exploring the relations between multiple pixels in a non-Euclidean space. Therefore, a graph convolutional network (GCN) is one of the effective models for aggregating information between HSI samples [24], [25], [26], [27], [28]. To fully exploit the ground-object information, Jia et al. [29] proposed a graph-in-GCN Jiang et al. [30] integrated graph structure learning into a multihead attention mechanism to implement a combination of transformer models and GCNs.

In the field of HSI classification, GCNs are able to aggregate information between samples, while CNNs excel in extracting spectral–spatial features. To exploit the advantages of both algorithms, Dong et al. [31] proposed a weighted feature fusion model that combines CNN and graph attention network to comprehensively explore the features of HSI data. In a similar vein, Zhou et al. [32] introduced an integrated network framework based on multihop graph [33], which effectively integrates multihop graph with a multiscale convolutional module for improved performance. It is worth noting that most GCNs feed all samples into the network during each forward propagation, which results in the trained network not being able to directly predict new samples. To address this limitation, Hong et al. [34] proposed a novel fusion network combining GCNs and CNNs. Inspired by the concept of minibatch learning in CNNs, a GCN subnetwork trained using minibatches is designed to enable correspondence between input graph samples and patch samples in CNNs. Liu et al. [35] designed a suitable batch processing network by integrating Euclidean and non-Euclidean neighbors when constructing graph samples.

These collaborative models, which incorporate both GCNs and CNNs, have demonstrated significant advances in classification accuracy. However, there are still some challenges in the feature fusion process. Different from CNNs in both feature extraction methods and output, GCNs obtain large-scale 2-D feature maps through matrix operations. Therefore, there is a certain degree of incompatibility between the feature maps of the two networks. In this case, direct fusion interferes with the extracted features from each other. In addition, unlike the multidimensional moving weighted summation employed in CNNs, the utilization of fixed nodes and 2-D matrix multiplication limit the feature extraction capability of graph convolutions to some extent. Therefore, these issues hinder the comprehensive and flexible exploration of the intricate spatial–spectral features in HSIs.

To overcome these difficulties, in this article, we propose a multichannel feature extraction module based on miniGCN, and a class-guided attention mechanism is designed to fuse the features extracted by CNNs and GCNs. Specifically, the main contributions of this article are given as follows.

1) To further enhance the integration of GCN and CNN features, a multilayer fusion framework is proposed in this article. This framework incorporates a class-guided attention mechanism and achieves the multiple fusions of GCN and CNN at different layers.
2) In this article, we propose a novel multichannel GCN (MGCN) module inspired by miniGCN. The adjacent matrices of MGCN aggregate the information between samples across different channels, enabling the propagation of 3-D feature maps within graph convolutional layers. Moreover, 3-D convolutional layers are used in MGCN to further enhance the compatibility of GCNs and CNN features during subsequent fusion.
3) We design a feature fusion strategy based on a class-guided attention mechanism. This approach guides the fusion of multiple MGCN modules by utilizing the attention vectors generated by the CNN modules. With the cooperation of CNNs and GCNs, the feature extraction capability of the proposed model is effectively improved.
4) In the process of constructing graph samples, we propose a straightforward yet effective data augmentation strategy for graph sampling. This approach generates additional samples by perturbing the position of target pixels within a certain region.

The rest of this article is organized as follows. Section II presents related work. Section III describes the proposed algorithm and framework. Section IV is devoted to the experiments and analysis of the results. Finally, Section V concludes this article.

II. Related Works

In this section, we present the architecture of GCNs and related applications in HSI classification tasks. Subsequently, we review some techniques for integrating features from CNNs and GCNs through fusion methods.

A. GCNs and Related Models

In the context of randomly selecting training samples, the distribution of labeled samples in HSIs conforms to a non-Euclidean structure. Therefore, GCNs are commonly employed in HSI classification [32], [36], [37]. The node information of the graph structure corresponds to the spectral data of the target pixels, while the edge information represents the similarity relations between pixels.

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The adjacent matrix plays a pivotal role in GCNs for extracting information between samples (edge information). The construction of the adjacent matrix $\mathbf{A}$ involves encoding the relationship between each sample and its neighbors. Within this matrix, each element $\mathbf{A}_{ij}$ indicates the strength of the association between the ith sample and the jth sample. Some works employ similarity of spectral feature [34] or spatial distance [37] to quantify the interrelation between distinct samples as follows:

$$\mathbf{A}_{ij} = \left(\sum_{E=1}^{E} \left| \mathbf{x}_{iE} - \mathbf{x}_{jE} \right|^{p} \right)^{\frac{1}{p}} \tag{1}$$

$$\mathbf{A}_{ij} = \exp\left(- \frac{\left\| \mathbf{v}_i - \mathbf{v}_j \right\|^2}{\sigma^2} \right) \tag{2}$$

where E denotes the dimension of vectors, p is an optional parameter, $\mathbf{x}_i$ represents the locations of the ith sample, $\mathbf{v}_i$ denotes the spectral feature of ith sample, and σ is a parameter to control the width of the radial basis function defined by (2).

The input data, which aggregate information between samples, is computed by matrix multiplication of adjacent matrices and node embedding vectors. Considering optimization strategies such as normalization [38], the final propagation process can be described as follows:

$$\mathbf{H}^{(l+1)} = f\left(\widetilde{\mathbf{D}}^{-\frac{1}{2}} \widetilde{\mathbf{A}} \widetilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(l)} \mathbf{W}^{(l)} + \mathbf{b}^{(l)} \right) \tag{3}$$

where $\mathbf{H}^{(l)}$ represents the output data of the lth layer, $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ represent the weights and biases of the lth layer, $\widetilde{\mathbf{A}}$ denotes $\mathbf{A}+\mathbf{I}$, $\mathbf{I}$ denotes the identity matrix, $\widetilde{\mathbf{D}}_{ii} = \Sigma_j \widetilde{\mathbf{A}}_{ij}$, and $f(\cdot)$ represents the activation function.

Analyzing (3) from a dimensional perspective, n denotes the total number of nodes in the HSI graph structure and k denotes the number of bands; then, $\mathbf{D}, \widetilde{\mathbf{D}}, \mathbf{A}, \widetilde{\mathbf{A}} \in \mathbb{R}^{n \times n}$; $\mathbf{b}^l$, $\mathbf{H}^l \in \mathbb{R}^{n \times k}$; and $\mathbf{W}^l \in \mathbb{R}^{k \times k}$. Some models [24], [31], [39] reduce the number of nodes in the graph structure by replacing individual pixels with superpixels. Zhou et al. [32] employed PCA as a preprocessing technique to efficiently degrade the bands and subsequently alleviate the burden of computational cost. MiniGCN [34] reduces the memory cost and enables the trained model to predict new samples. Chen et al. [40] proposed feature fusion via deep residual GCNs to exploit deeper features in HSIs.

Recent works have shown that GCN-related models can achieve HSI classification with high accuracy. Su et al. [41] integrated a cuckoo search strategy into a pretrained GCN to adaptively improve graph structure. Cai et al. [42] used patch features of HSI to construct graph samples and mined more discriminative features through a random walk transition probability matrix. In addition, similar to the multiscale form of patch samples, Xi et al. [43] proposed a cross-scale graph prototypical network. The network uses a multibranch framework and 1-D convolutions to achieve information fusion between neighborhoods of different scales. Yang et al. [26] built a small-scale graph by superpixel segmentation and metric learning to dynamically learn pixel representations and expanded it into a multiscale version to obtain richer spatial information.

Although GCN models achieve relatively high classification accuracy, most of them require high memory costs. Methods such as superpixel segmentation can reduce computation while losing details, and it is difficult to achieve a more adequate feature extraction such as the sliding window method for CNN models. Therefore, in this article, we propose a hybrid framework that jointly uses CNN and batch GCN to reduce the operation cost and extract more detailed information.

B. Feature Fusion of GCNs and CNNs

In hybrid networks of CNNs and GCNs, the issue of feature incompatibility needs to be considered when fusing intermediate features. Specifically, the features obtained by the CNN are 3-D data structures of multiple channels. It is derived from a sample of patches of fixed shape in Euclidean space. The convolution operation is implemented through a sliding window convolution kernel. There is a certain Euclidean spatial correspondence between feature maps and samples. GCN samples are constructed by selecting neighbors in a non-Euclidean space. The permutation of each neighbor node in the adjacency matrix is not specified in general, and there is no Euclidean space correspondence between its features and those of the CNNs. In addition, when constructing graph samples using superpixels, GCNs extract features at the superpixel level, while CNNs focus on pixel-level features [32]. Therefore, in the domain of HSI classification, several hybrid models with CNNs and GCNs fuse features in the final fully connected layer to avoid feature incompatibility [34], [44], [45]. This approach relies on the feature extraction ability of a single network (CNN or GCN). It is difficult to fully mobilize the cooperativity of the hybrid network for different modules. To further enhance collaboration and exploit intermediate features extracted by both GCNs and CNN modules, Zhou et al. [32] proposed a novel fusion strategy. First, the multiple features generated by the multihead self-attention mechanism [46] are concatenated according to the channel dimension. Second, the extracted graph structure features are returned to pixelwise feature form via the transformation matrix. Finally, the resulting feature maps of CNNs and GCNs are fed into the cross-attention fusion module. This transformation matrix is also frequently used in superpixel-based HSI classification tasks [31], [32]. However, the transformation matrix between pixel and superpixel features does not fully represent the transformation relationship between GCNs features and CNN features. We propose a fusion framework based on MGCN and guided attention mechanism (MG^{2}A), which implements learning of the transformation process and multilayer fusion to achieve more adequate collaboration between GCNs and CNNs.

III. Proposed Method

The flowchart of the proposed MG^{2}A is shown in Fig. 1. The framework is divided into three main parts. First, multichannel graph structure samples are constructed and data augmentation is performed in the preprocessing stage. Second, in the GCNs part, we extract HSI features with four MGCN modules. Finally, the proposed class-guided attention mechanism is used to fuse the features of CNNs and GCNs.

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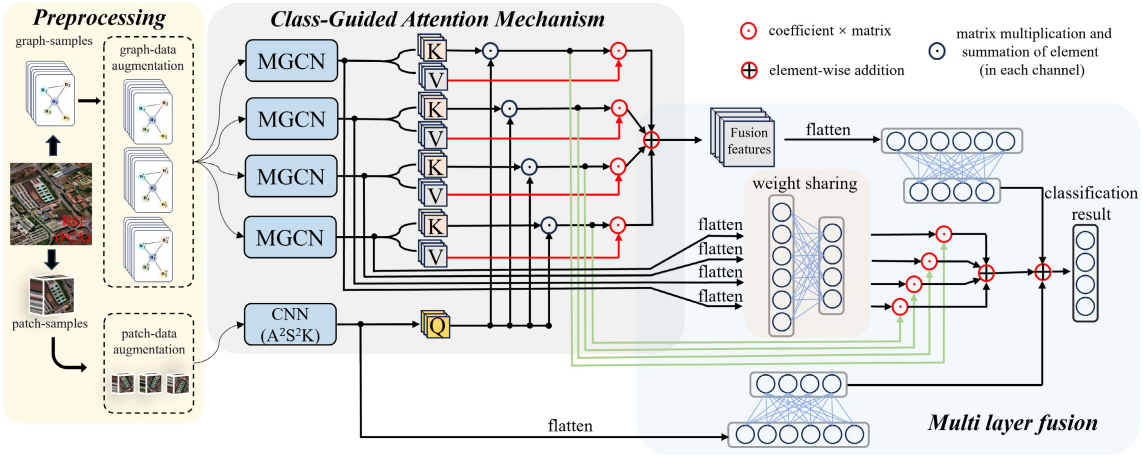


Fig. 1. Framework of MG^{2}A. The proposed model consists of a data augmentation process and a class-guided attention mechanism, a fusion strategy of multiple fully connected layers, where the class-guided attention mechanism consists of four MGCN modules and a CNN module.

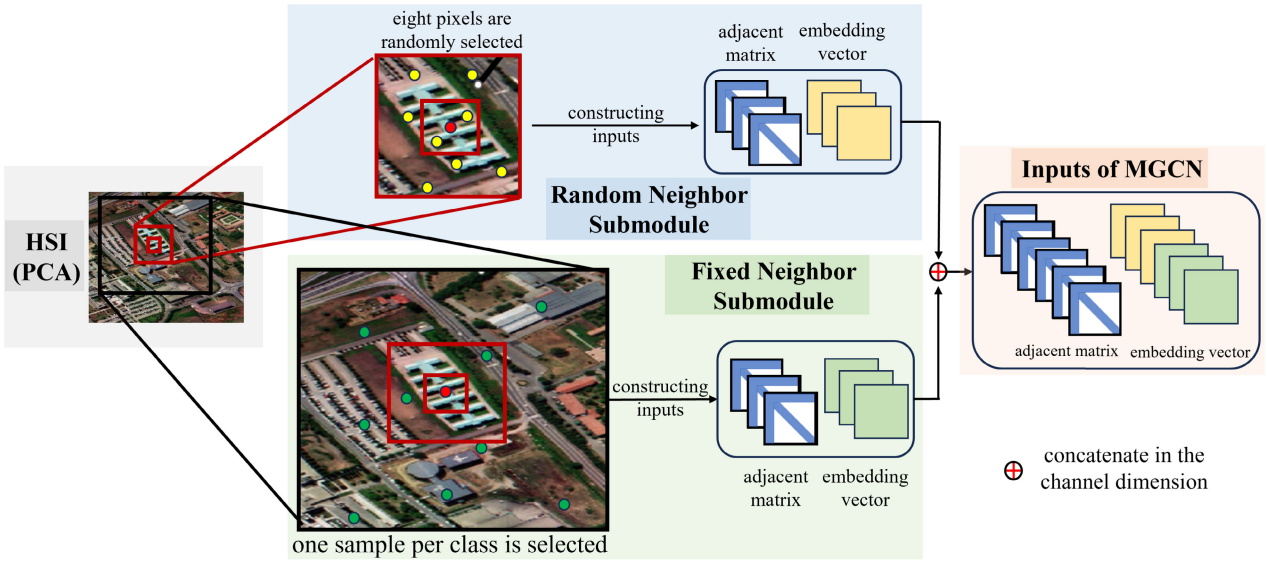


Fig. 2. Construction of graph samples. The red points represent the target pixels, and the yellow and green points represent the random and fixed neighbors of the target pixels, respectively.

A. Construction and Data Augmentation of Graph Samples

To match the minibatch training mode, it is necessary to construct a graph structure sample for each labeled target pixel in MGCN. In this article, h, w, and k denote the height, width, and number of retained principal components of the patch-form samples after PCA dimensionality reduction, respectively. $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denotes an undirected graph, where $\mathcal{V}$ and $\mathcal{E}$ denote the vertex and edge sets, respectively. The proposed MGCN module consists of six channels, including three channels for random neighbor submodules and three channels for fixed neighbor submodules. The two submodules are constructed in the same way, and the distinction lies in the choice of parameters and neighbors. As shown in Fig. 2, within the random neighbor submodule, we select neighboring nodes by randomly sampling from a rectangular region surrounding the target sample. The classes of these neighbor nodes are not considered in this procedure. The height and width of this region are set to $(3 \times h)$ and $(3 \times w)$, respectively. In this region, only n pixels with nonbackground labels are searched as neighbor nodes. However, not all labeled pixels have enough neighbor nodes. Compared to other labeled pixels with many neighbors, spatial features of such target pixels pay more attention to pixels with few neighbors. For these special samples, we repeatedly fill in a finite number of neighbor pixels until the n nodes that can be used to construct a graph structure are satisfied. To highlight the target pixel, only the edges of the nodes between the target pixel and each neighbor pixel are constructed in the graph sample. Thus, the adjacent matrix calculates data only in the first row, the first column, and the diagonal, and all other elements are zero. In both submodules, three aspects of the measurement are included: spatial position, spectral angle, and spectral vector difference. The adjacent matrix for each channel is computed as follows:

$$\mathbf{A}_{ij}^{r_1} = k^r \exp\left(-\frac{\|\mathbf{c}_i - \mathbf{c}_j\|}{\sigma^{r_1}}\right) \tag{4}$$

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$$\mathbf{A}_{ij}^{r_2} = k^r \exp\left(-\frac{\cos^{-1}\left(\mathbf{v}_i^T \mathbf{v}_j\right)}{\left(\mathbf{v}_i^T \mathbf{v}_i \cdot \mathbf{v}_j^T \mathbf{v}_j\right)^{\frac{1}{2}} \sigma^{r_2}}\right) \tag{5}$$

$$\mathbf{A}_{ij}^{r_3} = k^r \exp\left(-\frac{\left\|\mathbf{v}_i - \mathbf{v}_j\right\|^2}{\sigma^{r_3}}\right) \tag{6}$$

where $\mathbf{A}_{ij}^{r_1}$, $\mathbf{A}_{ij}^{r_2}$, and $\mathbf{A}_{ij}^{r_3}$ denote the elements of the three adjacent matrices from different channels; k^r denotes the weight of the random neighbor submodule; and r_1, r_2, and r_3 represent the channels of spatial position, spectral angle, and spectral vector difference in random neighbor submodule, respectively. $\mathbf{c}_i$ and $\mathbf{c}_j$ denote the spatial positions of the ith and jth pixels, respectively, while $\mathbf{v}_i$ and $\mathbf{v}_j$ denote the spectral features of the ith and jth pixels, respectively. σ controls the range of values for the different measures. The weight of each channel is determined by k^r and σ together.

The strategy for selecting neighbors of the fixed neighbor submodule is different from the above. The main difference is that the random neighbor submodule randomly selects neighbor nodes in the construction process of each graph sample, while the fixed neighbor submodule uses the same set of neighbor nodes in the construction of all graph samples. To maximize the utilization of HSI information, it is crucial to select neighbors from different classes as extensively as possible. As shown in the following equation, after determining the neighbor nodes, the adjacent matrices of the fixed neighbor submodule are computed in a similar way:

$$\mathbf{A}_{ij}^{f_1} = k^f \exp\left(-\frac{\left\|\mathbf{c}_i - \mathbf{c}_j\right\|}{\sigma^{f_1}}\right) \tag{7}$$

$$\mathbf{A}_{ij}^{f_2} = k^f \exp\left(-\frac{\cos^{-1}\left(\mathbf{v}_i^T \mathbf{v}_j\right)}{\left(\mathbf{v}_i^T \mathbf{v}_i \cdot \mathbf{v}_j^T \mathbf{v}_j\right)^{\frac{1}{2}} \sigma^{f_2}}\right) \tag{8}$$

$$\mathbf{A}_{ij}^{f_3} = k^f \exp\left(-\frac{\left\|\mathbf{v}_i - \mathbf{v}_j\right\|^2}{\sigma^{f_3}}\right) \tag{9}$$

where f_1, f_2, and f_3 represent the channels of spatial position, spectral angle, and spectral vector difference, respectively, in the fixed neighbor submodule. By concatenating the adjacent matrices of the two submodules at the channel dimension, we construct graph samples with six distinct channels as input for MGCN.

In MG^{2}A, we perform data augmentation on the patch-form samples of the CNN module and the graph-form samples of the MGCN module, respectively. For patch-form samples, we use vertical and horizontal flips to increase the training samples by a factor of two. For the graph-form samples, we perturb the selected target pixel locations. As shown in Fig. 3, the spectral features of this target sample are replaced by those of a surrounding pixel when constructing the adjacent matrix of the additional training samples. To avoid constructing mislabeled samples, we limit the range of coordinate variation to two pixels. This data augmentation strategy is based on the following three points. First, HSI classification belongs to the pixelwise classification task, where pixels of the same class object are concentrated. Second, hyperspectral remote sensing images suffer from low spatial resolution due to limitations in imaging equipment, resulting in an endmember

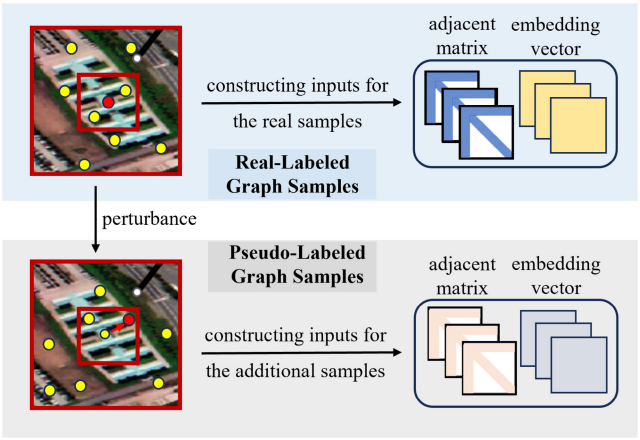


Fig. 3. Proposed graph sample data augmentation method. The red arrow points from the original target pixel to the perturbed target pixel. In this process, all random neighbors are reselected, and the fixed neighbors are unchanged.

mixing problem. Consequently, even if pixels are selected at the edges of the ground object, the spectral information from neighboring pixels still maintains a certain degree of similarity. Finally, there are two channels in the graph-form samples for extracting spatial location features. Since only pixel-level position perturbations are generated, these features and labels still maintain high confidence. Based on this, adjusting the weight coefficient k of the adjacent matrix corresponding to the spatial location can increase the confidence of the label of the new sample.

B. MGCN

Before stating the propagation pattern of the MGCN, the structure of the data involved in the operation needs to be clarified. The input to the MGCN is a multichannel graph sample consisting of m adjacent matrices. Each graph sample contains n nodes, and the node signals are the k principal components after PCA. The propagation process of graph convolutional layers in MGCN is computed by

$$\mathbf{H}^{*(l+1)} = f\left(\widetilde{\mathbf{D}^*}^{-\frac{1}{2}} \widetilde{\mathbf{A}^*} \widetilde{\mathbf{D}^*}^{-\frac{1}{2}} \mathbf{H}^{*(l)} \mathbf{W}^{*(l)} + \mathbf{b}^{*(l)}\right) \tag{10}$$

where $\widetilde{\mathbf{D}^*}$, $\widetilde{\mathbf{A}^*} \in \mathbb{R}^{m \times n \times n}$ denote the degree matrix and the adjacent matrix, respectively, $\mathbf{b}^{*(l)}$, $\mathbf{H}^{*(l)} \in \mathbb{R}^{m \times n \times k}$ denote the bias and the feature map of the lth layer, and $\mathbf{W}^{*(l)} \in \mathbb{R}^{m \times k \times k}$ denotes the learnable parameters. Since the number of channels remains constant during propagation, the operation at each layer can be viewed as concatenation in the channel dimension after m 2-D matrix operations.

In the feature space, each channel of the sample updates the feature information of the nodes according to the graph convolution. The adjacency matrix is kept unchanged in each subsequent graph convolutional layer, and only the node signal features are updated. In the feature space of the lth layer graph convolution of MGCN, we stack the updated feature information of each channel (the feature graph in a matrix form of a single channel). The activation function is performed, and new 3-D feature maps with the same input size are obtained.

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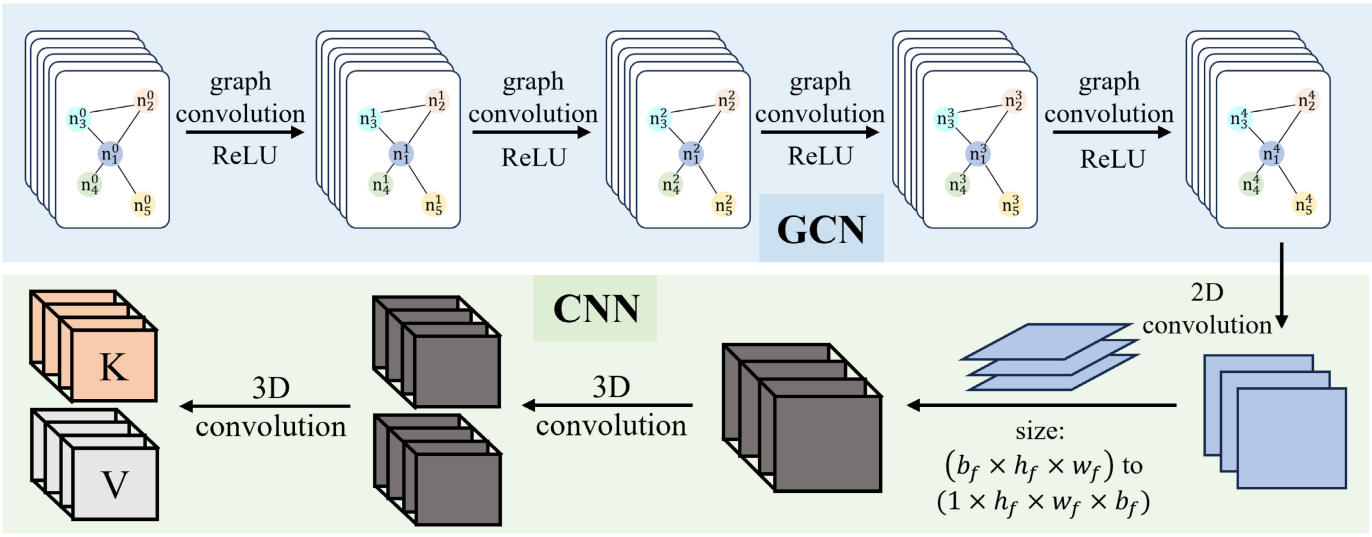


Fig. 4. Structure diagram of the MGCN. The model consists of four graph convolutional layers, one 2-D convolutional layer, and two 3-D convolutional layers. The output of the model is two 3-D feature maps (**Q** and **K**) for the class-guided attention mechanism. b_f, h_f, and w_f denote the number of channels, length, and width of feature maps in the 2-D convolution, respectively.

The feature map and the adjacency matrix are fed into the $(l+1)$th layer, and the above procedure is repeated to achieve node information update and propagation in the feature space.

The structure of the MGCN is shown in Fig. 4. The 3-D features of the multichannel graph samples are extracted by the graph convolution algorithm described above. Since the features are derived from graph structure samples, they need to be transformed before fusion with CNN features to improve compatibility. However, it is difficult to mathematically obtain the exact solution of the transformation process. Therefore, we embed the 3-D convolution operation into the MGCN, which learns this feature transformation process with the help of the network. Specifically, first, the 3-D feature maps output by the four graph convolutional layers are fed into a 2-D convolutional layer and a 3-D convolutional layer. Second, the convolution results are used for subsequent fusion with the feature maps of the CNN module. Finally, the classification results are used as supervision signals during training so that the feature maps extracted by MGCN are progressively more adaptable to the feature maps of CNN modules.

C. Class-Guided Attention Mechanism

Inspired by Wang et al. [24], we design a class-guided attention mechanism based on the multihead attention mechanism for fusion. The mechanism consists of multiple MGCN modules and a single CNN module. As shown in Fig. 1, each MGCN module outputs the 3-D feature maps of **K** (key) and **V** (value) through a 3-D convolutional layer, and the CNN outputs a single-channel feature map **Q** (query) of the same size. The core of class-guided attention is that it can adaptively assign different weights (obtained by **K** and **Q** calculation) to the features of multiple MGCN modules for fusion according to the CNN features of different categories. The formula is shown as follows:

$$\mathbf{H}_{\text{out}} = \sum_i \mathbf{C}_i \mathbf{V}_i \tag{11}$$

$$\mathbf{C}_i = \text{Softmax}\left(f_{\text{sum}}\left(\sum_j \mathbf{K}_{ij}\mathbf{Q}_{ij}\right)\right)\mathbf{V}_i \tag{12}$$

where $\mathbf{H}_{\text{out}}$ denotes the 3-D feature map of the output of the class-guided attention mechanism, $\mathbf{V}_i$ and $\mathbf{K}_i$ are the outputs of the ith MGCN module, $\mathbf{C}_i$ is the coefficient of $\mathbf{V}_i$, $\mathbf{K}_{ij}$ is the matrix of the jth channel in $\mathbf{K}_i$, $\mathbf{Q}_{ij}$ is the jth channel matrix corresponding to $\mathbf{K}_{ij}$ and generated by the CNN module, and $f_{\text{sum}}(\cdot)$ represents the sum of all the elements of the matrix. Specifically, the **Q** obtained by the CNN module is multiplied with the **K** generated by the MGCN module on each channel separately. All the elements in each matrix result are then summed separately and fed into a Softmax function to serve as weight coefficients for **V** of the current MGCN. The procedure is similar to convolving **K** with a convolution kernel (**Q**) of the same size and obtaining the coefficients. However, the difference is that this process is performed by matrix multiplication. Finally, the fusion results are obtained by adding the output feature maps **V** of multiple MGCN models according to the group weights.

Similar to the use of multiple convolutional kernels in CNN, multiple MGCN modules with different learnable parameters can extract HSI features more fully. To better collaborate with multiple MGCN modules, we develop a class-guided attention mechanism that enables the network to independently select the weights of each MGCN module for fusion. During this process, **Q** of the CNN module represents the features of the current sample. With the help of the feature compatibility adjustment of the MGCN module, **K** is used to fuse with **Q** and output the attention coefficients of each MGCN module. Finally, the attention coefficient vector is multiplied with **V** to obtain the MGCN fusion result. Since the feature maps of the CNN module are used to guide the output feature maps of the GCNs by computing attention coefficients for indirect fusion, this further improves the compatibility between GCN features and CNN features. It is worth noting that the main

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task of the CNN module lies in generating **Q**, which contains the features of the target samples. Therefore, this mechanism has no special requirements on the structure of CNNs. Only the output of the model needs to be adjusted to support matrix multiplication with **K**. In this article, we adopt A^2S^2K as the CNN module of MG^2A.

K, **V**, and **Q** belong to different network characteristics. Considering the possible incompatibility of the two features, 2-D convolution and 3-D convolution are added to the MGCN module. These convolutional layers can not only further mine the information in the graph features but also be considered as learning the transformation process between two different features. This makes the feature maps of **K**, **V**, and **Q** more compatible. In order for **Q** to express information about its own class, we employ the quadratic constraint. That is, **Q** is flattened and fed into fully connected layers to participate in fusion and classification. Under the supervision of the class label, **Q** can be further made to have class features.

Moreover, the samples in MGCN correspond to CNN samples one by one. Both samples consist of the target pixel and its neighboring pixels. This implies that the sources of information for **Q** and **V** correspond to each other. Based on this, the compatibility between the two features is further improved by the feature transformation process in MGCN and the backpropagation of the overall framework. The output of the CNN can be used as a query term **Q** of the MGCN with good compatibility under the cooperation of similar information sources and feature transformation.

D. Multilayer Fusion Framework

The proposed multilayer fusion framework is shown in Fig. 1, which includes two fully connected fusions in addition to the fusion in the MGCN module. First, we leverage the attention coefficients in the class-guided attention mechanism, which further fuses the results of the fully connected layers (with shared parameters) of different MGCN modules. Second, the fusion results of CNN and MGCN are fed into fully connected layers separately. Finally, the results of the three fully connected layers are summed to obtain the final classification result.

IV. Experiments

A. Dataset Description

In this article, we evaluate the performance of MG^2A from multiple perspectives using the Pavia University (PU) dataset, the Salinas Valley (SV) dataset, the Indian Pines (IP) dataset, and the GRSS-DFC-2013-Houston (HS) dataset.

1) The PU dataset was captured by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor at the PU. The dataset consists of 610 × 340 pixels with nine labeled classes of ground-based objects. The spatial resolution is 1.3 meters per pixel (mpp), and the spectral resolution is 3.74 nm. The spectral range is from 430 to 860 nm with 115 bands. After removing the interference bands, 103 bands remain.
2) The SV dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the SV, CA, USA. The scene consists of 512 × 217 pixels with 16 labeled ground object categories. The spatial resolution is 3.7 mpp. Considering the interference bands, 204 spectral channels are eventually retained.
3) The IP dataset is a scene from the northwest Indiana region, USA, captured by the AVIRIS Sensor. The dataset consists of 145 × 145 pixels, covering 16 classes of labeled samples, with a spatial resolution of 20 mpp, a spectral resolution of 10 nm, and a wavelength range of 400–2500 nm, totaling 220 spectral bands. Considering noise and water absorption bands, 200 spectral channels are eventually retained.
4) The HS dataset was collected by the National Center for Airborne Laser Mapping at the University of Houston in June 2012 and was distributed for the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Image Analysis and DataFusion (IADF) Contest [47]. The dataset contains 349 × 1905 pixels, covers 15 categories and a wavelength range of 0.38–1.05 mum, and has a spatial resolution of 2.5 m.

Table I lists the pseudocolor composite images and the number of labeled samples in each class for the above four datasets.

B. Experimental Setup

Our proposed framework contains four MGCN modules, a CNN module, and a fusion strategy based on a class-guided attention mechanism. A^2S^2K is used as the CNN module. To match A^2S^2K with the class-guided attention mechanism, we discard the fully connected layers of the model and cascade the input of the residual blocks of the last two layers, which are fed into the 3-D convolutional layers to obtain Q. The convolutional operations of all layers use the padding technique to make the feature map size of each channel the same as the input size of the model, and all convolution kernels have an edge length of 3. In the experiments, we use the Adam optimizer with a minibatch size of 128, and the learning rate is set to 0.001. The number of epochs is 150 and an early stopping strategy is used. Both the patch size and the number of nodes of the graph samples are set to 9.

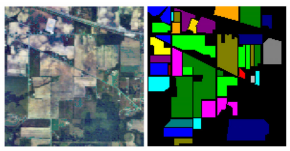
We selected ten HSI classification models for comparative analysis, including SVM, DBDA [15], SSCNN [19], A^2S^2K [20], P-RESNET [21], miniGCN [34], Multiple Vision Architecture-Based Hybrid Network (MVAHN) [36], mixed residual convolutions with Vision Transformer model (MRViT) [48], dual-view spectral and global spatial feature fusion (DSGSF) network [49], and attention multihop graph and multiscale convolutional fusion network (AMGCFN) [32]. Both MVAHN and MRViT incorporate transformer modules in addition to convolution operations. MVAHN combines GCNs, CNNs, and vision transformer models to explore HSI features using multiple types of networks. MRViT mitigates the accuracy degradation by incorporating multiple residual blocks before the vision transformer module.

To fairly compare the classification performance, all models adopted the hyperparameters recommended in the original article. The experiments were repeated independently five times for each model, and the results were averaged. The

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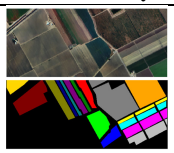
TABLE I
IP, SV, PU, AND GRSS-DFC-2013-HOUSTON DATASETS

Indian Pines



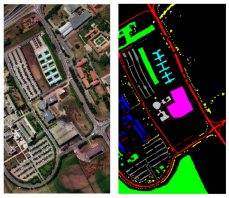
No.	Color	Class name	Samples	Proportion
C1		Alfalfa	46	0.45%
C2		Corn-no till	1428	13.93%
C3		Corn-min till	830	8.10%
C4		Corn	237	2.31%
C5		Grass/Pasture	483	4.71%
C6		Grass/Trees	730	7.12%
C7		Grass/pasture-mowed	28	0.27%
C8		Hay-windrowed	478	4.66%
C9		Oats	20	0.20%
C10		Soybeans-no till	972	9.48%
C11		Soybeans-min till	2455	23.95%
C12		Soybeans-clean till	593	5.79%
C13		Wheat	205	2.00%
C14		Woods	1265	12.34%
C15		Bldg-Grass-Tree-Drives	386	3.77%
C16		Stone-steel towers	93	0.91%
-		Background	10776	-

Salinas Valley



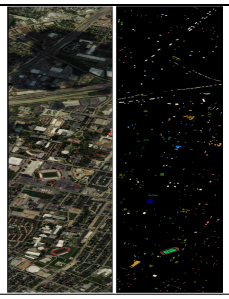
No.	Color	Class name	Samples	Proportion
C1		Brocoli_green_weeds 1	2009	3.71%
C2		Brocoli_green_weeds 2	3726	6.88%
C3		Fallow	1976	3.65%
C4		Fallow_rough_plow	1394	2.58%
C5		Fallow_smooth	2678	4.95%
C6		Stubble	3959	7.31%
C7		Celery	3579	6.61%
C8		Grapes_untrained	11271	20.82%
C9		Soil_vinyard_develop	6203	11.46%
C10		Corn_senesced_weeds	3278	6.06%
C11		Lettuce_romaine_4wk	1068	1.97%
C12		Lettuce_romaine_5wk	1927	3.56%
C13		Lettuce_romaine_6wk	916	1.69%
C14		Lettuce_romaine_7wk	1070	1.98%
C15		Vinyard_untrained	7268	13.43%
C16		Vinyard_vertical_trellis	1807	3.34%
-		Background	56975	-

Pavia University



No.	Color	Class name	Samples	Proportion
C1		Asphalt	6631	15.50%
C2		Meadows	18649	43.60%
C3		Painted metal sheets	2099	4.91%
C4		Gravel	3064	7.16%
C5		Trees	1345	3.14%
C6		Shadow	5029	11.76%
C7		Bare Soil	1330	3.11%
C8		Bitumen	3682	8.61%
C9		Bricks	947	2.21%
-		Background	164624	-

GRSS-DFC-2013-Houston



No.	Color	Class name	Samples	Proportion
C1		Healthy grass	1251	8.32%
C2		Stressed grass	1254	8.34%
C3		Synthetic grass	697	4.64%
C4		Trees	1244	8.28%
C5		Soil	1242	8.26%
C6		Water	325	2.16%
C7		Residential	1268	8.44%
C8		Commercial	1244	8.28%
C9		Road	1252	8.33%
C10		Highway	1227	8.16%
C11		Railway	1235	8.22%
C12		Parking Lot 1	1233	8.20%
C13		Parking Lot 2	469	3.12%
C14		Tennis Court	428	2.85%
C15		Running Track	660	4.39%
-		Background	649816	-

overall accuracy (OA), average accuracy (AA), Kappa coefficient (Kappa), and accuracy per class are used to compare and evaluate the classification performance of each model; 100 training samples per class were selected for the PU, SV, and IP datasets, and 50 training samples per class were selected for the HS dataset. When the total number of samples for a certain class does not reach the required number of training samples, half of the samples are randomly selected as training samples. The validation set samples are obtained by randomly sampling among all samples, with the same total number as the training set. The remaining labeled samples in the dataset form the test set, which is used to evaluate the model performance. Considering that the proposed method uses random flipping as the data augmentation strategy, the same strategy is applied to the models with patch-form inputs. In addition, we conduct ablation experiments and analyze the effectiveness of each module in the proposed framework.

C. Experimental Result

Tables II–V report the classification results of each model on the four datasets. Overall, SVM suffers from salt-and-pepper noise on all four datasets, and there is a significant accuracy gap between SVM and other deep learning models. This means that it is difficult to exploit all the features of the HSI data using only spectral information, which directly affects the classification results. Our proposed model achieves the best OA, AA, and Kappa on all four datasets and the best accuracy on most classes. This validates the advanced classification performance of our designed framework.

The classification results for the PU dataset are shown in Table II. SVM uses only spectral information in the classification process. Therefore, the accuracy comparison between SVM and other models can reflect the degree of response of different classes to spectral and spatial information. The SVM algorithm can achieve significantly better classification performance in class 9 (shadows), indicating that the class is more sensitive to spectral features. MG^{2}A achieves the best accuracy score in this class. The results demonstrate that the model is able to extract spectral features more comprehensively. For class 1 (asphalt), which has the lowest classification accuracy of SVM, MG^{2}A can still achieve the best classification performance among all models. This proves that the proposed framework can simultaneously achieve excellent spatial and spectral feature extraction capabilities. Moreover, A^{2}S^{2}K shows competitive classification performance with respect to other 3D-CNN models, which is the main reason why we use it as the CNN module of MG^{2}A. MiniGCN also shows excellent classification performance, validating the effectiveness of batch learning GCNs for HSI feature extraction. In addition, AMGCFN shows obvious oversmoothing while achieving high accuracy. This may be due to the fact that graph models aggregate features in a way that relies more on spatial information when the entire graph is used as input. MRViT is ranked among the top in all evaluation metrics and may benefit from the introduction of location information and the multihead attention mechanism of the transform model. Correspondingly, we design a class-guided attention mechanism in MG^{2}A and introduce relative location information by fixing the neighbor submodule during graph sample construction. Experimental results show that MG^{2}A can effectively coordinate the various modules and algorithms in the framework.

The classification results for the SV dataset are shown in Table III. The classification performance of SVM using only spectral information on the dataset shows a clear disadvantage. AA of A^{2}S^{2}K, DBDA, P-RESNET, miniGCN, and MRViT all exceed 98%. This suggests that better classification accuracy for SV datasets requires the ability of the model to extract spatial features. A possible reason is that the spatial distribution of labeled ground objects in this dataset is relatively more regular, and the clustering properties of similar ground objects are more pronounced. Furthermore, from the analysis of the results, class 15 (Vinyard-untrained) samples are more difficult to identify and have the lowest classification accuracy among these models. However, the accuracy of our proposed model can reach 95.76% for such a class. Thus, MG^{2}A is relatively more balanced in its ability to recognize ground objects of different classes in the HSI high-precision classification task.

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TABLE II

QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS IN TERMS OF OA, AA, AND KAPPA ON THE PU DATASET. THE BEST ONE IS SHOWN IN BOLD

Class	SVM	A^2S^2K	SSCNN	DBDA	P-RESNET	miniGCN	MVAHN	MRViT	DSGSF	AMGCFN	MG^2A
1	82.59	98.00	97.80	**99.40**	97.64	98.13	98.68	98.81	99.09	97.93	98.68
2	89.82	99.86	99.57	99.46	99.54	99.61	99.82	99.78	99.59	98.35	**99.94**
3	84.04	95.07	90.15	**99.77**	92.63	95.22	96.61	96.05	94.42	99.56	99.08
4	96.32	97.70	97.88	97.02	97.68	98.01	96.85	97.65	98.83	97.26	**99.31**
5	99.36	99.73	99.95	99.92	99.88	99.69	99.51	99.87	97.72	**100.00**	**100.00**
6	88.25	96.77	96.19	98.28	97.21	97.31	99.34	98.96	96.78	**100.00**	**100.00**
7	95.04	99.18	86.40	97.80	97.94	96.67	95.23	96.53	96.89	**100.00**	99.83
8	86.54	90.92	88.68	92.96	88.10	93.60	92.85	93.63	94.99	**99.70**	99.20
9	99.88	97.23	91.07	94.91	97.78	96.24	85.42	98.01	99.02	97.61	**100.00**
OA (%)	89.05	97.95	96.74	98.44	97.39	98.13	98.40	98.53	98.32	98.66	**99.60**
AA (%)	91.32	97.16	94.19	97.73	96.49	97.27	97.24	97.70	97.48	98.93	**99.56**
Kappa	0.8563	0.9727	0.9566	0.9791	0.9652	0.9750	0.9787	0.9804	97.76	98.22	**0.9947**

TABLE III

QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS IN TERMS OF OA, AA, AND KAPPA ON THE SV DATASET. THE BEST ONE IS SHOWN IN BOLD

Class	SVM	A^2S^2K	SSCNN	DBDA	P-RESNET	miniGCN	MVAHN	MRViT	DSGSF	AMGCFN	MG^2A
1	99.63	97.66	99.89	**100.00**	99.98	99.93	**100.00**	99.97	99.95	**100.00**	99.95
2	99.53	**100.00**	99.99	**100.00**	**100.00**	99.99	99.99	99.96	99.83	**100.00**	**100.00**
3	99.63	99.51	99.93	99.98	99.83	99.76	99.75	**100.00**	97.86	**100.00**	**100.00**
4	99.77	99.40	97.54	95.71	97.25	96.65	98.91	**99.88**	97.12	99.75	99.76
5	96.51	99.86	99.41	**99.92**	99.85	99.84	99.87	99.41	99.55	99.57	99.52
6	99.53	**100.00**	**100.00**	**100.00**	99.86	99.98	**100.00**	**100.00**	99.87	99.99	**100.00**
7	99.37	99.98	**100.00**	99.97	99.79	99.95	99.99	**100.00**	99.68	99.95	**100.00**
8	81.18	96.58	95.01	99.00	95.26	96.10	96.59	96.22	**99.62**	96.15	98.67
9	97.51	99.81	99.69	99.70	99.87	99.75	99.75	99.98	99.60	**100.00**	**100.00**
10	93.61	99.38	99.04	**99.49**	99.14	99.45	98.88	99.18	94.68	99.18	98.89
11	95.35	99.85	98.52	99.89	99.94	99.20	99.93	98.57	96.06	**100.00**	99.68
12	99.89	99.94	99.73	**100.00**	99.63	99.67	99.97	99.89	99.66	**100.00**	**100.00**
13	98.28	99.87	**100.00**	99.92	98.59	99.80	99.94	**100.00**	99.23	**100.00**	**100.00**
14	94.54	99.94	98.23	99.65	98.56	98.66	99.97	99.09	96.75	99.35	**100.00**
15	69.00	89.45	77.13	90.43	88.21	84.50	90.04	**99.53**	95.04	99.39	99.09
16	99.41	**100.00**	**100.00**	**100.00**	99.14	**100.00**	**100.00**	**100.00**	97.66	**100.00**	**100.00**
OA (%)	90.54	97.66	95.05	98.18	97.05	96.62	97.72	98.96	98.39	99.01	**99.49**
AA (%)	95.17	98.97	97.76	98.98	98.43	98.33	98.97	99.34	98.26	99.58	**99.72**
Kappa	0.8945	0.9739	0.9448	0.9797	0.9671	0.9623	0.9746	0.9884	98.20	98.90	**0.9943**

TABLE IV

QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS IN TERMS OF OA, AA, AND KAPPA ON THE IP DATASET. THE BEST ONE IS SHOWN IN BOLD

class	SVM	A^2S^2K	RSSN	DBDA	P-RESNET	MINIGCN	MVAHN	MRViT	DSGSF	AMGCFN	MG^2A
1	82.43	90.48	92.88	96.49	**100.00**	**100.00**	89.37	99.13	**100.00**	**100.00**	92.11
2	76.69	97.61	90.71	97.01	98.10	98.02	98.69	96.47	98.65	96.64	**99.36**
3	76.51	96.09	87.01	96.05	93.47	93.70	94.20	95.92	**98.41**	97.49	96.10
4	88.74	93.16	87.46	97.43	97.17	94.48	93.14	92.85	81.25	94.43	**98.36**
5	93.38	**100.00**	96.84	99.01	98.14	98.57	98.12	99.30	98.59	98.79	99.52
6	91.19	98.18	96.96	98.05	98.50	97.71	93.18	**99.31**	96.30	98.05	97.65
7	86.11	**100.00**	**100.00**	99.17	85.03	97.50	98.67	96.47	86.67	**100.00**	**100.00**
8	97.06	**100.00**	**100.00**	99.98	**100.00**	99.95	99.55	99.55	**100.00**	99.61	**100.00**
9	**100.00**	37.57	**96.97**	84.43	77.49	53.30	**100.00**	87.47	**100.00**	**100.00**	**100.00**
10	81.24	93.98	87.14	95.99	94.93	92.93	90.83	89.40	90.45	91.26	**96.98**
11	68.87	97.68	95.63	96.75	98.67	97.83	98.83	99.11	**99.28**	99.25	98.25
12	79.80	96.29	88.52	96.76	93.03	94.43	98.01	97.15	95.85	**98.90**	98.18
13	97.68	90.08	97.94	96.74	98.29	96.57	99.19	**99.73**	99.01	98.33	95.24
14	92.93	98.74	96.17	99.12	98.88	98.76	99.68	99.69	**99.91**	99.76	99.27
15	71.48	95.59	92.19	96.70	95.80	95.18	94.37	91.38	95.29	94.26	**96.10**
16	93.00	52.64	83.08	82.37	85.39	67.09	92.96	**96.54**	84.91	84.62	86.75
OA (%)	80.56	96.61	92.88	97.13	97.19	96.56	96.87	97.01	97.32	97.47	**98.11**
AA (%)	86.07	89.88	93.09	95.75	94.56	92.25	96.17	96.22	95.29	96.96	**97.12**
Kappa	0.7803	0.9611	0.9179	0.9669	0.9675	0.9605	0.9641	0.9657	0.9689	0.9709	**0.9781**

The classification results for the IP dataset are shown in Table IV. The OA achieved by SVM on this dataset is only 80%, and both the AA and Kappa values fall below the OA values. In contrast, the OA, AA, and Kappa of other models incorporating spatial information exceed 90%. This discrepancy may also be attributed to the intrinsic characteristics

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TABLE V
QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS IN TERMS OF OA, AA, AND KAPPA ON THE HS DATASET. THE BEST ONE IS SHOWN IN BOLD

	SVM	A^2S^2K	RSSN	DBDA	P-RESNET	MINIGCN	MVAHN	VIT	DSGSF	AMGCFN	MG^2A
1	93.84	98.76	93.88	96.76	97.89	96.02	**99.37**	98.48	97.33	96.61	98.87
2	98.75	**100.00**	98.41	99.26	99.56	99.58	99.91	99.41	98.96	98.63	99.69
3	99.38	**100.00**	99.66	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	99.56	99.90	99.93
4	92.46	97.67	92.81	97.30	97.81	97.83	96.92	**99.83**	98.54	98.98	98.17
5	97.65	94.04	91.54	**100.00**	95.22	96.75	95.89	**100.00**	97.85	99.93	99.91
6	97.82	98.86	86.36	97.74	98.49	91.49	99.24	98.71	98.12	99.14	**99.84**
7	90.07	98.52	90.80	98.26	94.01	93.32	**99.02**	93.43	95.74	96.05	97.95
8	54.19	98.49	86.85	99.80	98.43	95.50	**100.00**	94.67	97.59	91.29	94.88
9	76.62	92.07	84.18	91.84	92.34	91.24	96.59	96.05	96.73	94.23	**98.19**
10	78.93	94.40	81.66	95.24	92.61	90.93	99.12	99.66	98.08	99.22	**99.46**
11	63.46	98.65	83.38	98.84	98.64	96.86	99.01	98.38	98.19	98.53	**99.57**
12	54.35	98.80	90.23	94.70	92.95	93.14	93.89	98.12	97.94	97.57	**99.03**
13	49.40	92.68	82.35	88.74	92.22	92.37	91.18	97.76	95.76	98.06	**99.18**
14	97.09	97.57	88.74	99.72	**100.00**	99.67	97.57	99.75	98.23	99.88	99.97
15	98.52	**100.00**	95.74	99.83	98.14	98.97	96.35	**100.00**	99.22	**100.00**	**100.00**
OA (%)	81.66	97.24	89.78	97.18	96.23	95.43	97.73	98.07	97.79	97.49	**98.80**
AA (%)	82.84	97.37	89.77	97.20	96.55	95.58	97.60	98.28	97.86	97.87	**98.95**
Kappa	0.8016	0.9702	0.8895	0.9695	0.9592	0.9506	0.9755	0.9791	0.9761	0.9729	**0.9870**

of the dataset. Similar to the SV dataset, this dataset has a more regular distribution of ground-based objects. The experimental results further reflect the advantage of efficient extraction of deep spatial features for HSI classification accuracy improvement. While some very advanced deep learning models (DBDA and MRViT) can achieve an OA of 97% on this dataset in our experiments, our proposed framework is able to improve the accuracy by about 1% on top of this, resulting in an OA of over 98%. AA and Kappa are also significantly improved.

The classification results for the HS dataset are shown in Table V. Both DSGSF and SVM achieve more unambiguous boundary classification results. However, the SVM still suffers from severe salt-and-pepper noise and, thus, has the lowest classification accuracy. AMGCFN, DBDA, and A^2S^2K can achieve an OA of 97%, but there is clear boundary ambiguity in the classification results. This discrepancy in the bounds may be caused by the different dependence of the different models on the spatial and spectral information. In particular, for models with patch samples as input, boundary samples suffer from the problem of having a large number of overlapping neighborhoods, and thus, it is difficult to distinguish the classes of boundary pixels. In addition, the distribution of labeled objects in the HS dataset is more scattered. Compared to SV and IP datasets, the HS dataset has more diverse and complex spatial features. Our proposed framework is able to achieve optimal classification accuracy with relatively modest boundary classification. This shows that MG^2A can effectively balance the spectral and spatial information contributions to the classification results and achieve the best classification accuracy.

Figs. 5–8 visually show the classification results of the different models on the four datasets, respectively. The SVM using only spectral information exhibits severe salt-and-pepper noise phenomena. The classification results of SSCNN, P-RESNET, DBDA, and A^2S^2K with integrated spatial information are smoother, but the accuracy of bound pixels decreases. This is because a large amount of spatial information leads to information leakage [50] while improving classification accuracy, which makes the contextual information of different pixels near the boundary excessively similar.

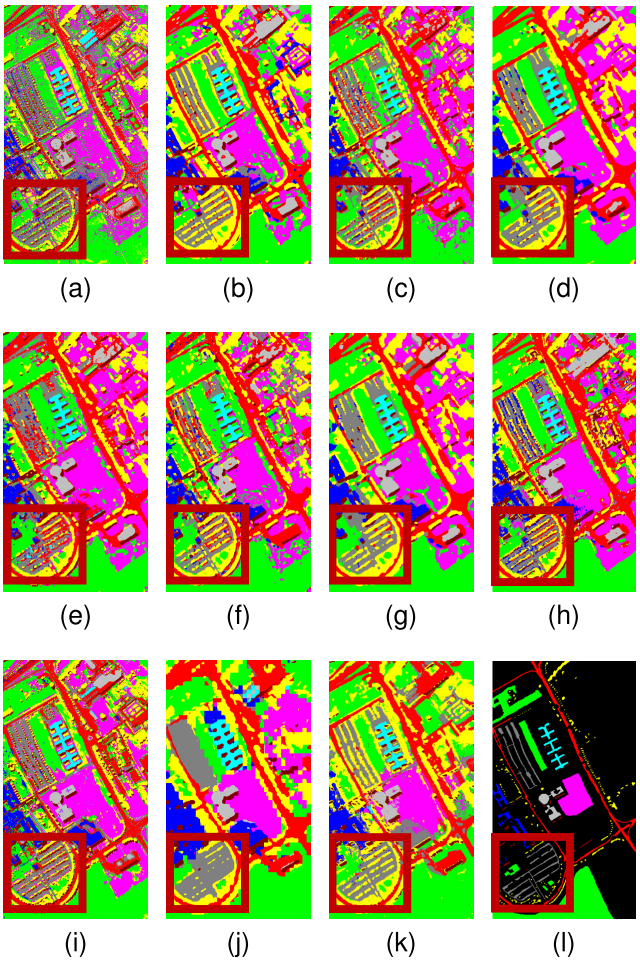


Fig. 5. Classification maps of different methods for the PU dataset. (a) SVM. (b) A^2S^2K. (c) SSCNN. (d) DBDA. (e) P-RESNET. (f) miniGCN. (g) MVAHN. (h) MRViT. (i) DSGSF. (j) AMGCFN. (k) MG^2A. (l) Ground truth.

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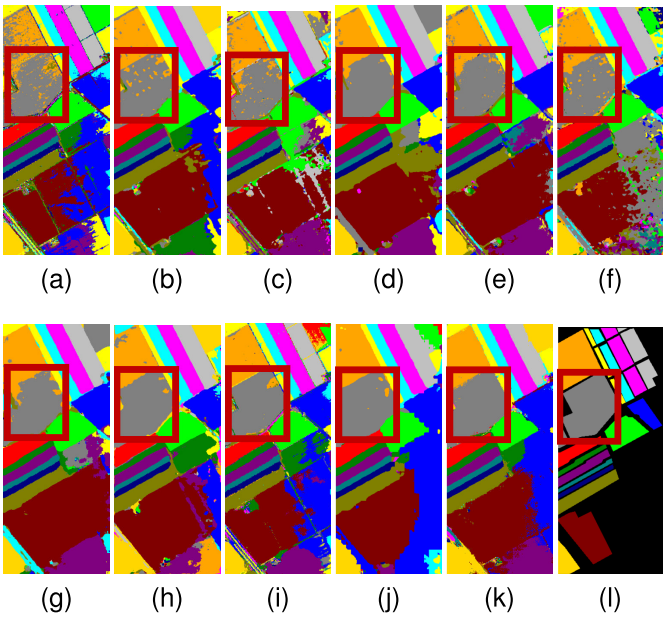


Fig. 6. Classification maps of different methods for the SV dataset. (a) SVM. (b) A^2S^2K. (c) SSCNN. (d) DBDA. (e) P-RESNET. (f) miniGCN. (g) MVAHN. (h) MRViT. (i) DSGSF. (j) AMGCFN. (k) MG^2A. (l) Ground truth.

Moreover, it is difficult to take into account the relationship between surrounding and target pixels in non-European space. Although DSGSF achieves high-precision classification, its entire-batch input approach leads to high memory costs. AMGCFN, MVAHN, and MRViT further smooth the classification results. This may be caused by the introduction of location information, which makes features of concentrated samples from the same class easier to recognize. However, misclassified pixels are more concentrated, which decreases the classification performance. Due to the incompatibility between CNN intermediate features and GCN intermediate features, miniGCN and MVAHN only convert these two features into vectors for fusion. Our proposed model not only contains the deep 3D-CNN module but also adds the MGCN module, which contains both spatial position and information between samples. With the cooperation of both modules, the best classification results are achieved. In addition, as shown in Fig. 9, to compare the proposed framework more fully with other state-of-the-art models, we choose different numbers of training samples and conduct experiments on four datasets. MG^2A achieves the best OA rate on all four datasets, which verifies the competitive classification performance of the MG^2A model. It indicates that our proposed method can fully coordinate GCN and CNN models and effectively exploit intermediate features.

D. Ablation Experiment

To evaluate the effectiveness of each module, we design ablation experiments. The experiment analyzes the relevant modules by retaining different components in the proposed framework. In the experiments, we consider the fixed neighbor submodule, random neighbor submodule, CNN module (including some adjustments to A^2S^2K for adaptation to MGCN), four MGCN modules directly additive fusion, and guided fusion and MG^2A without 2-D and 3-D convolution layers in MGCN. The experimental results are shown in Table VI. The random neighbor submodule and the fixed neighbor submodule have better classification accuracy on different classes, respectively. MG^2A combined with the two submodules can fully exploit their respective strengths and achieve the best classification performance. In experiments where only MGCN is retained in the model, the fusion strategy using the class-guided attention mechanism outperforms the direct additive fusion strategy by 0.47% in OA. Moreover, the classification accuracy of multiple MGCN models is better than that of a single MGCN model. These experimental phenomena demonstrate that the fusion strategy based on the class-guided attention mechanism can make the intermediate feature maps of GCNs and CNNs more compatible, thus improving the classification performance of the model. To better evaluate the performance of MG^2A, we remove the 2-D and 3-D convolutional layers in MGCN in the ablation experiments. Compared to MG^2A, the experimental results are reduced by only about 0.5%. This suggests that the excellent classification accuracy of MG^2A is more likely due to the multichannel graph convolution rather than the 2-D and 3-D convolution operations in MGCN. At the same time, this result validates the effectiveness of our introduction of 2-D and 3-D convolutions to mitigate feature compatibility. Finally, the CNN module alone cannot achieve the best classification accuracy, which is improved by 1.24% after adding MGCN. The results further show that the cooperation of MGCN and CNN can more fully extract the HSI features and, thus, improve the recognition ability of the model for the target object.

E. Parameter Discussion

In the data augmentation process, the perturbation strength of the original sample determines the confidence of the label of the new sample, which affects the classification performance of the model. This experiment discusses the relationship between edge length and classification accuracy for different perturbation ranges. To further focus on the effectiveness of the graph data augmentation strategy, we only retain four MGCN modules (additive fusion) for experiments on four datasets. The experimental results are shown in Fig. 10(a), where the classification accuracy shows a trend of first increasing and then decreasing with increasing perturbation range. This is because expanding the perturbation range increases the difference between the new sample and the original sample, and expanding the perturbation range beyond a certain scale may decrease the confidence of the labels and, hence, the classification accuracy. This result is optimal when the perturbation range is moderate (the edge length of the perturbation range is two pixels). This range can take into account both the differentiation and confidence of the new samples.

To compare the efficiency of batch-based deep learning models, we calculate the efficiency performance of some related models. Table VII statistics the training time for each epoch, the prediction time for all test samples, and the total number of learnable parameters. The batch size for these models is 100 samples. The results show that both the time

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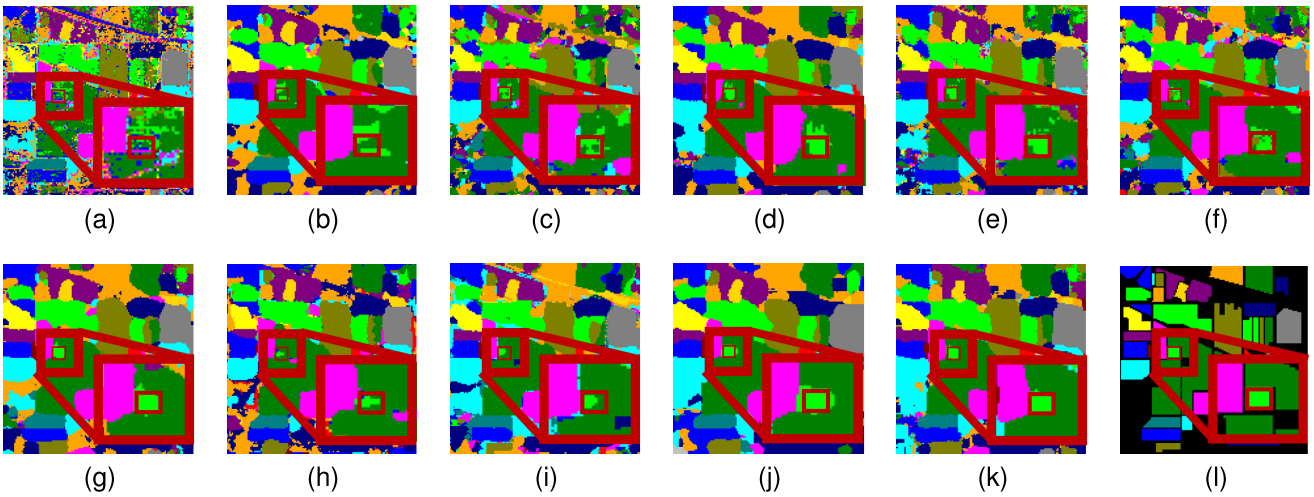


Fig. 7. Classification maps of different methods for the IP dataset. (a) SVM. (b) A^2S^2K. (c) SSCNN. (d) DBDA. (e) P-RESNET. (f) miniGCN. (g) MVAHN. (h) MRViT. (i) DSGSF. (j) AMGCFN. (k) MG^2A. (l) Ground truth.

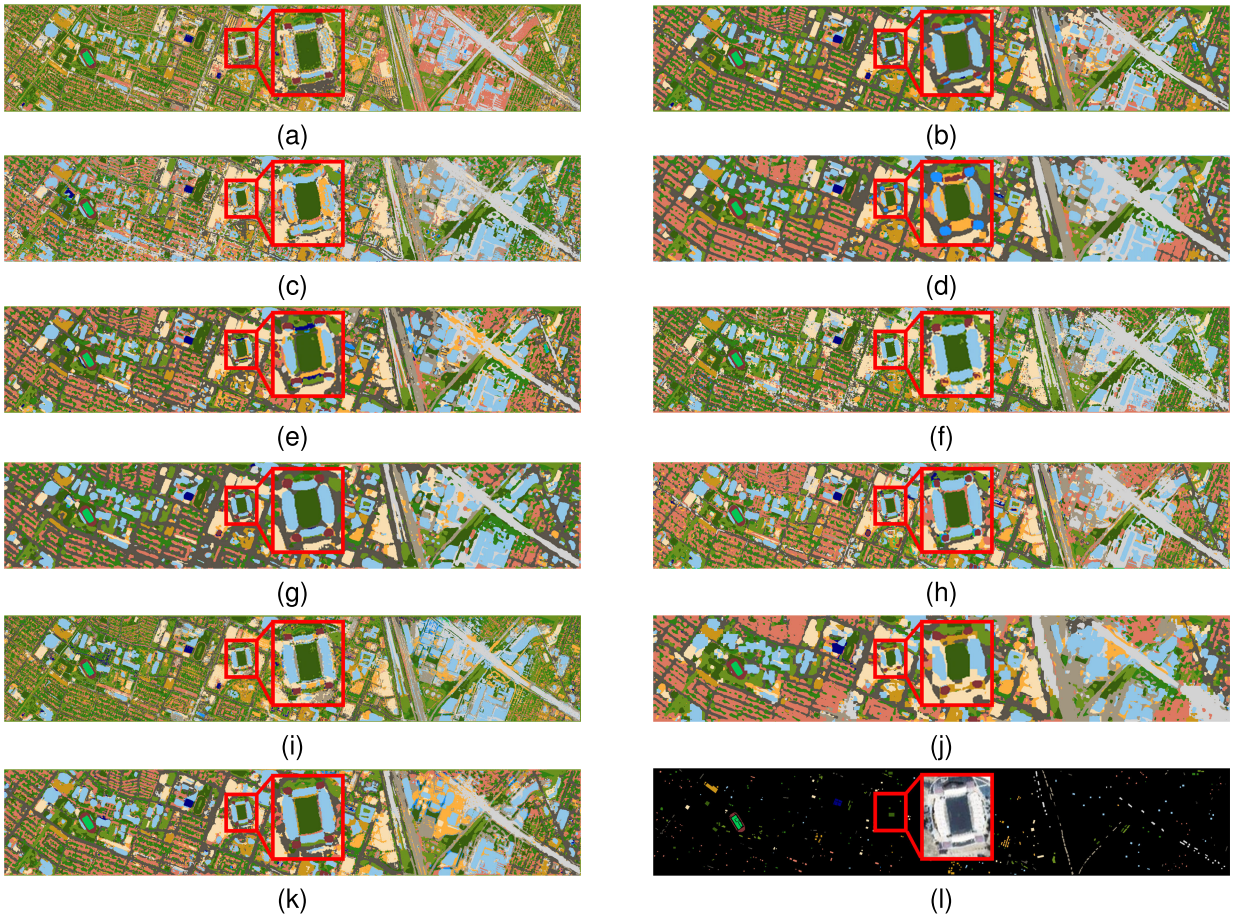


Fig. 8. Classification maps of different methods for the HS dataset. (a) SVM. (b) A^2S^2K. (c) SSCNN. (d) DBDA. (e) P-RESNET. (f) miniGCN. (g) MVAHN. (h) MRViT. (i) DSGSF. (j) AMGCFN. (k) MG^2A. (l) Ground truth.

cost and the number of parameters of MG^2A are moderate and within acceptable ranges compared to other networks.

The number of neighbor nodes of a graph sample is also one of the important parameters that affect the performance of the model. As shown in Fig. 10(b), the PU and HS datasets are urban scenes with relatively complex spatial features. In both datasets, the number of nodes changes causes stochastic fluctuations in model accuracy. For the SV and IP datasets, the distribution of ground objects is relatively more concentrated, and the accuracy shows an increasing trend with the number of nodes. This phenomenon may be due to the fact that the abundant number of nodes in SV and IP datasets favors the utilization of spatial information. However, PU and HS ground-based objects are relatively scattered, and it may be

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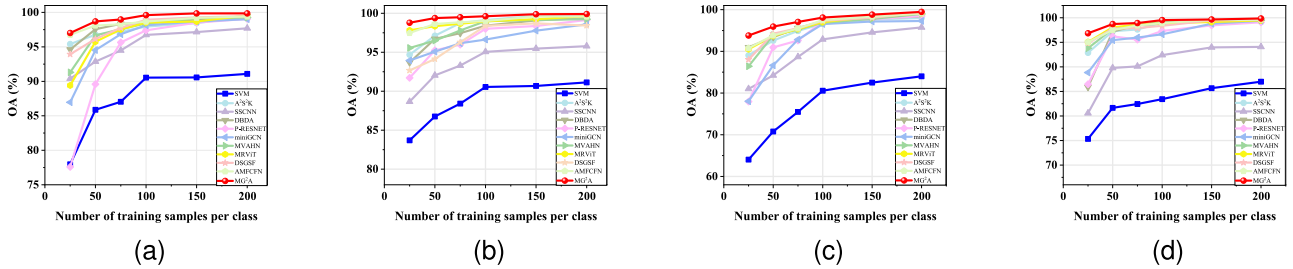


Fig. 9. Classification accuracy of different methods under different scales of training samples. (a) PU. (b) SV. (c) IP. (d) HS.

TABLE VI
OA OF DIFFERENT MODULES IN TERMS OF OA, AA, AND KAPPA ON THE PU DATASET. THE BEST ONE IS SHOWN IN BOLD

Class	Without Fixed Neighbor Channels	Without Random Neighbor Channels	CNN Module	Four MGCN (Guided)	Four MGCN (Sum)	MGCN	MG^2A without CNN in MGCN	MG^2A
1	**99.59**	99.52	99.58	98.70	99.32	98.46	98.21	98.68
2	99.33	99.36	99.44	99.59	99.24	99.35	99.68	**99.94**
3	96.82	98.20	97.53	95.28	94.89	96.18	**99.39**	99.08
4	98.74	98.84	98.40	95.18	89.24	88.54	97.35	**99.31**
5	98.93	99.92	99.10	99.99	**100.00**	99.79	99.92	**100.00**
6	97.72	99.54	96.47	98.55	98.67	96.64	99.23	**100.00**
7	99.34	98.25	99.05	95.94	98.42	98.41	99.25	**99.83**
8	96.77	96.49	94.51	94.91	94.78	92.12	98.15	**99.20**
9	94.75	99.76	94.75	95.24	93.27	92.72	99.88	**100.00**
OA (%)	98.67	99.05	98.36	98.17	97.70	97.08	99.09	**99.60**
AA (%)	98.00	98.88	97.65	97.04	96.43	95.80	99.01	**99.56**
Kappa	0.9822	0.9873	0.9781	0.9757	0.9693	0.9611	98.78	**0.9947**

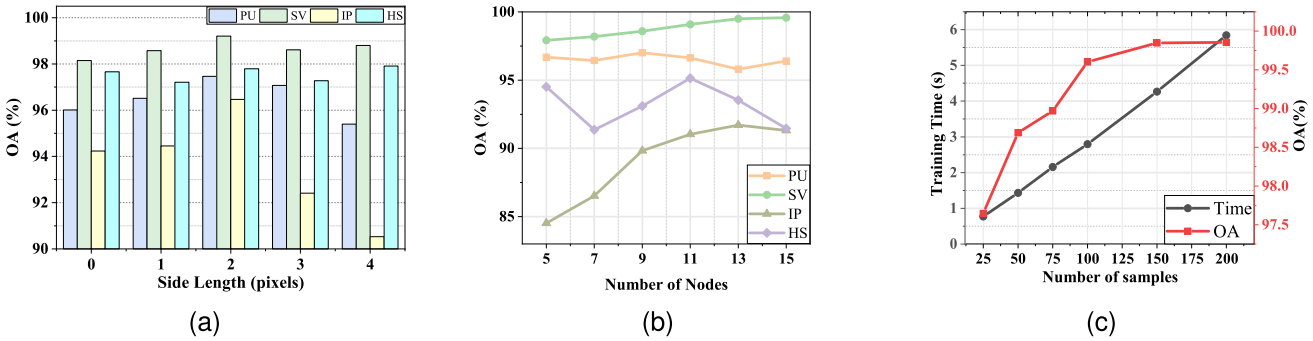


Fig. 10. Influence of parameters on model performance. (a) Disturbance strength. (b) Number of nodes. (c) Number of samples.

TABLE VII
COMPARISON OF RUNNING EFFICIENCY FOR DIFFERENT METHODS

Training Time per Epoch (s)				Test Time (s)	Params (K)
Dataset	PU	SV	IP	PU	
A^2S^2K	7.77	25.77	20.27	13.16	221.98
SSCNN	0.26	0.74	0.55	3.08	4869.48
DBDA	6.65	13.09	10.22	16.12	202.75
P-RESNET	1.34	2.72	2.09	12.23	1110.17
miniGCN	0.47	1.24	0.72	8.59	146.50
MVAHN	0.72	1.41	1.13	4.23	445.77
MG^2A	2.77	4.85	4.01	7.06	981.05

TABLE VIII
INFLUENCE OF THE NUMBER OF MGCNs ON OA

MGCNs	PU			SV		
	OA	AA	Kappa	OA	AA	Kappa
2	96.82	95.93	95.77	98.81	99.06	98.67
4	**98.69**	**98.41**	**98.26**	**99.39**	99.47	**99.32**
6	97.19	96.24	96.27	99.22	**99.52**	99.13
8	96.56	96.21	95.44	98.37	99.00	98.18

difficult for the model to significantly improve spatial feature extraction by adding more nodes.

In addition, another important hyperparameter in MG^2A is the number of MGCNs. We conduct the parameter experiment on the PU dataset and the SV dataset. As shown in Table VIII, MG^2A achieves the best performance on the PU dataset with four MGCNs and the best OA and Kappa on the SV dataset. Experiments show that the potential of MA2A can be further developed with an appropriate number of MGCNs.

To further discuss the time cost of MG^2A, we test its training time and classification accuracy using different numbers of training samples on the PU dataset. As shown in Fig. 10(c), similar to OA, the training time also shows a positive trend as the number of training samples increases. When using 100 training samples per class, the increase in classification accuracy is significantly reduced, and the training time per epoch does not exceed 3 s. MG^2A has a moderate time cost compared to other models. This is partly due to MG^2A using fewer neighboring nodes, PCA, and batch processing.

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V. Conclusion

CNNs have powerful feature extraction capabilities, GCNs can aggregate information between samples, and the collaborative efforts of CNNs and GCNs can be used to achieve HSI high-precision classification. Considering the incompatibility between feature maps of two different types of networks, we propose a multilayer fusion framework based on MG^{2}A. This framework fuses the features of CNNs and GCNs at multiple layers to explore the spectral–spatial information of HSIs more fully. As the core component of the proposed framework, MGCN implements two key functions. First, the module extracts HSIs features more comprehensively by introducing multiple information between samples. Second, MGCN learns the transformation process between GCN and CNN features by training the network, which effectively improves the compatibility between two different types of feature maps. Moreover, in this article, we design a fusion strategy based on the class-guided attention mechanism, which not only improves the feature extraction capability of the MGCN module but also achieves the fusion of intermediate features between the CNN module and the MGCN module. Experiments on four widely used HSI datasets show that our proposed MG^{2}A achieves higher classification accuracy compared to other state-of-the-art deep learning models.

Since the CNN features in the class-guided attention mechanism are mainly used to generate attention coefficients without directly changing the feature map V of the GCNs, this strategy improves the compatibility between different feature maps to some extent. Therefore, in future research, we plan to improve this strategy to extend its range of applications. Specifically, we consider using this strategy to guide multiscale features fusion, multimodal features fusion, and fusion between vision transformer features and CNN features to more fully exploit the rich spatial–spectral information contained in HSIs.

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